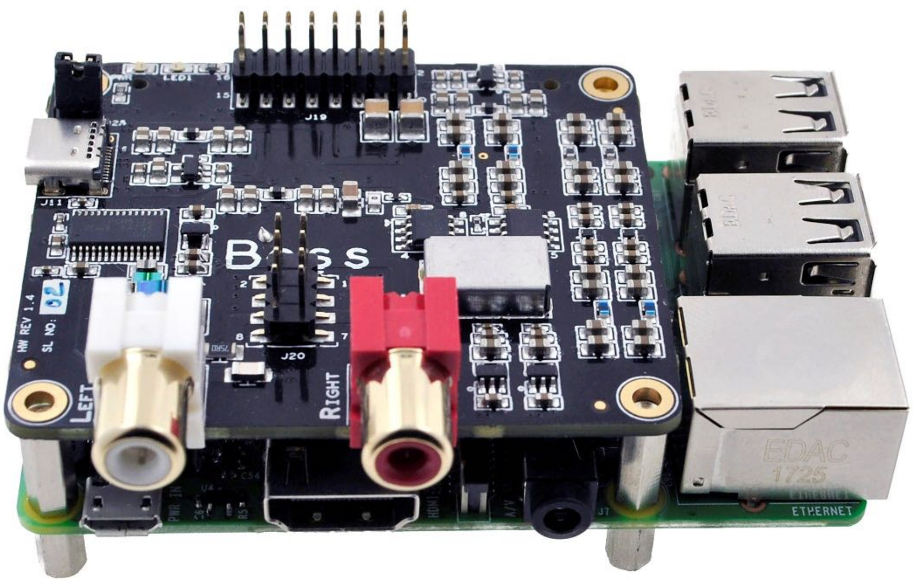
BOSS

TECHNICAL

MANUAL

BOSS DAC

BOSS DAC Shield has been designed for those seeking audio perfection. It comes with built-in Dual Master Clock Oscillators to support 44.1K & 48K series sampling rate audio playback. These are specially designed sound cards - compatible with RPI- 2 & 3 versions.



BOSS DAC Features:

- Dedicated 384 kHz/32bit high-quality DAC PCM5122 for best sound quality
- Audio output connectors: 2 x RCA (Left & Right) & ALLO Volt Amp header
- DAC SNR is 112dB
- DAC THD+N @ - 1dBFS are –93dB
- Full Scale Output of DAC is 2.1Vrms
- Dynamic Range of DAC is 112dB
- Sampling Frequency ranges from 8 kHz to 384 kHz
- Ultra-low-noise voltage regulators & LPF for optimal audio performance
- Automatically switching frequencies according to the input I2S signals
- Dual low jitter NDK crystal oscillators for Master Clock generation.

Operating Temperature:	0°C to 70°C
Board Size:	LWH = 67.4mm * 65mm * 22.2mm
Board Weight:	28g

BOSS DAC is a fully HAT-size add-on sound card for RPI- 2 & 3 version SBCs.

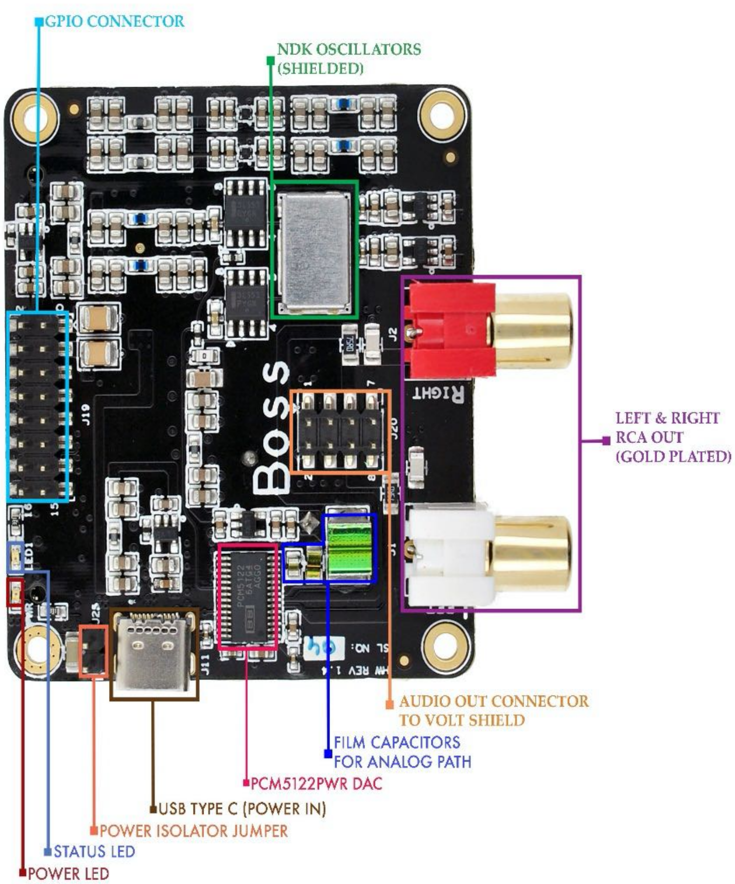
The BOSS will work on I2S Master Mode and Slave mode through the onboard Texas Instruments PCM5122 DAC IC. Master clock auto switching will take care by PCM5122 DAC and generates sample rate based I2S clocks out to RPI.

By using 45.1584/49.1520 MHz Ultra Low Phase Noise Oscillators, DAC generates bit perfect I2S clocks to RPI. This delivers excellent quality audio out through the BOSS RCA connectors.

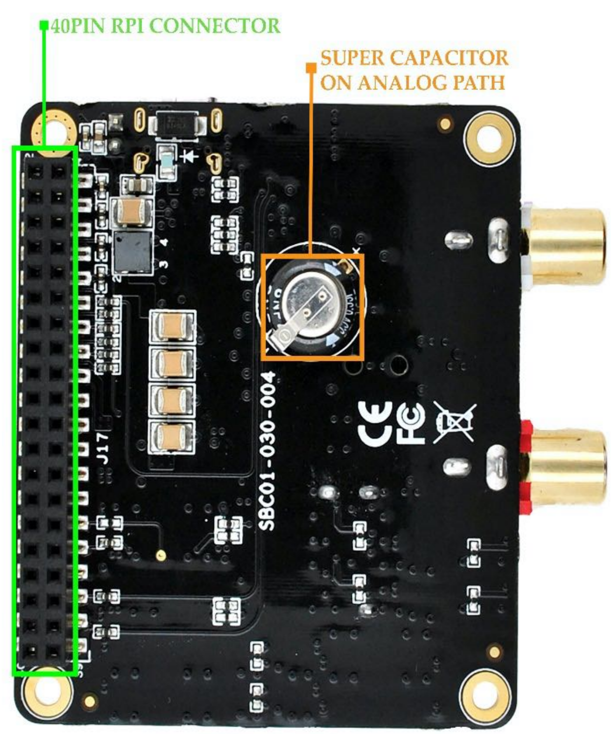
The BOSS can be connected to the RPI compatible 40-way header without any additional cable or soldering.

Component selection, Digital-Analog Partition and track layout have been in the forefront of our design to ensure noise immunity and best possible audio playback with the BOSS. The Analog Power section is designed with film capacitors and super Capacitor to achieve pure analog power to DAC.

TOP VIEW OF BOSS



BOTTOM VIEW OF BOSS



LED STATUS

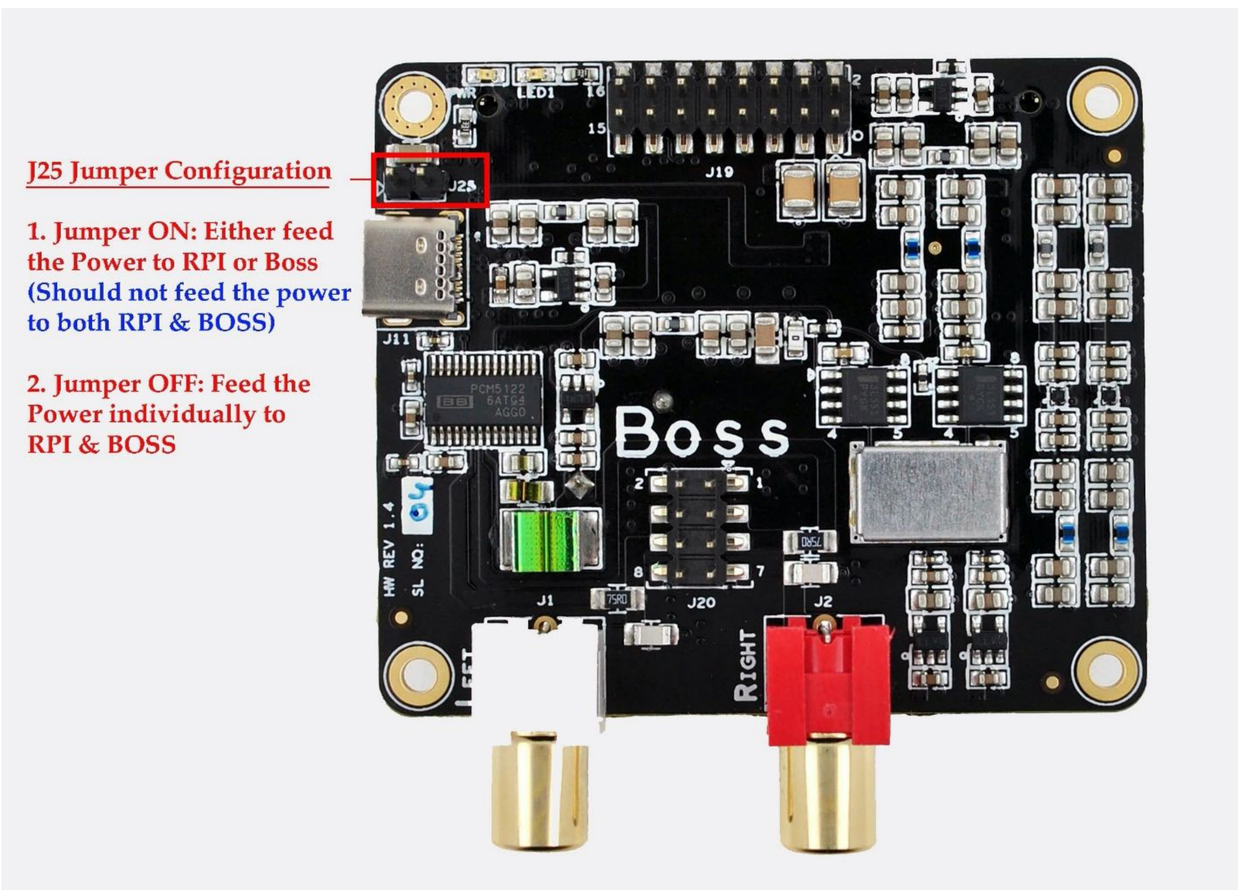
Green PWR LED: Power Indication

Green LED1: Play status

Power Options (5V):

1) J25 Jumper ON: Either feed the power to RPI or BOSS, should not feed the power to both RPI &BOSS
2) J25 Jumper OFF: feed the Power Individually to RPI &

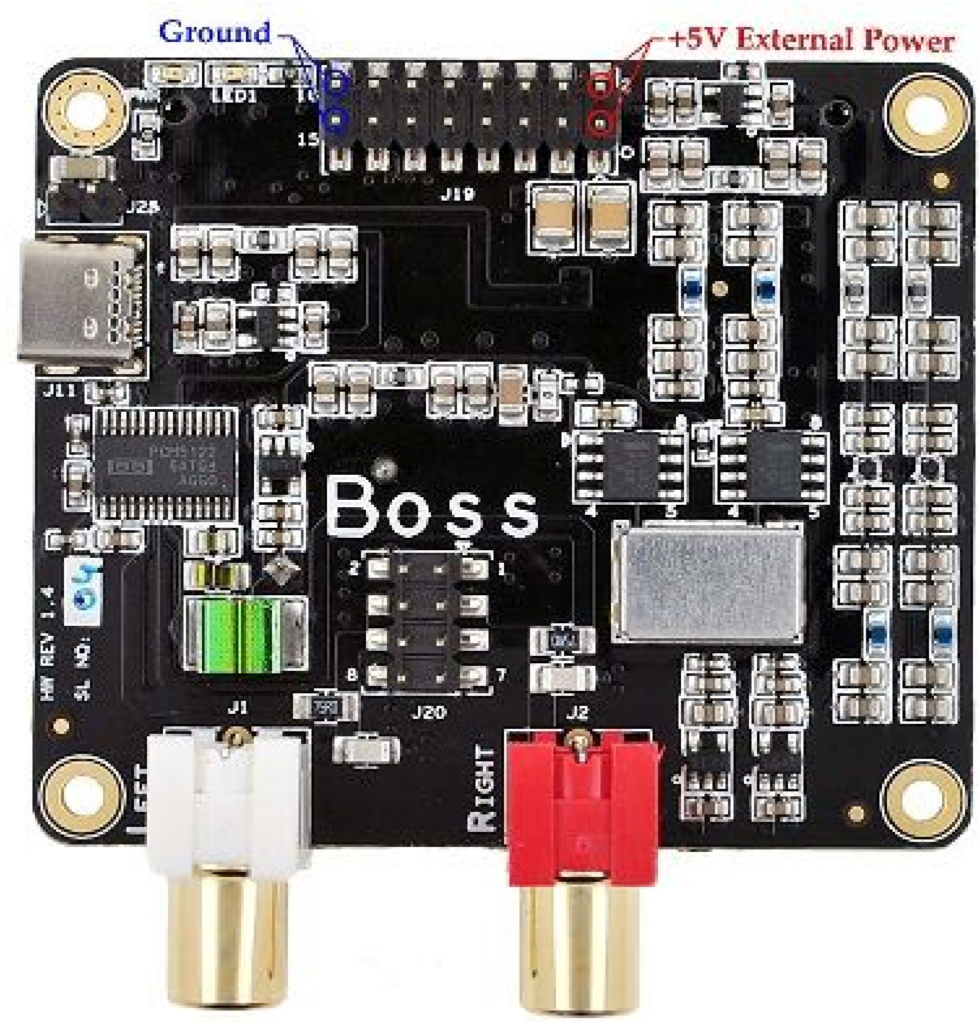
BOSS JUMPER SETTINGS



BOSS GPIO POWER PINS

P GND: J19(15) or J19(16)

+5V: J19(1)   or J19 (2)



BOSS HEADER PIN OUT DETAILS

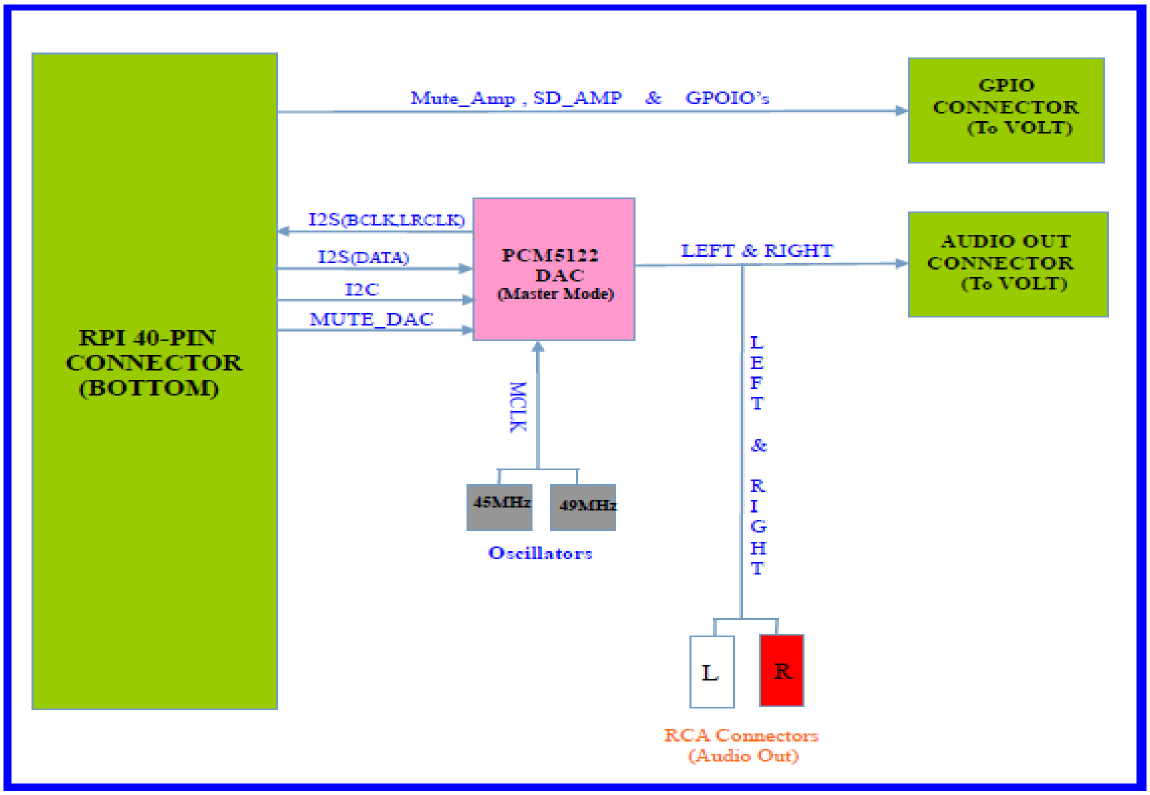
RPI	PIN	PIN	RPI
3V3	1	2	**DC +5V**
SDA1-I2C	3	4	**DC +5V**
SCL1-I2C	5	6	**GND**
GPIO4	7	8	**UART_TX**
GND	9	10	**UART_RX**
GPIO17	11	12	**I2S_BCLK**
GPIO27	13	14	**GND**
GPIO22	15	16	**GPIO23**
3V3	17	18	**GPIO24**
SPI_MOSI	19	20	**GND**
SPI_MISO	21	22	**GPIO25**
SPI_CLK	23	24	**GPIO8**
GND	25	26	**GPIO7**
ID_SD	27	28	**ID_SC**
GPIO5	29	30	**GND**
GPIO6/DMUTE	31	32	**GPIO12**
GPIO13	33	34	**GND**
I2S_LRCLK	35	36	**GPIO16**
GPIO26	37	38	**I2S_DIN**
GND	39	40	**I2S_DOUT**

** highlighted signals are used by BOSS board

BOSS J19 PIN OUT DETAILS (ALLO VOLT Header)			
RPI	PIN	PIN	RPI
5V	2	1	5V
3V3	4	3	3V3
GPIO26	6	5	SDA1-I2C
GPIO17	8	7	SCL1-I2C
GPIO27	10	9	NC
GPIO24	12	11	NC
GPIO22 / MUTE_AMP	14	13	GPIO23/ SDZ_AMP
GND	16	15	GND

BOSS J20 PIN OUT DETAILS (ALLO VOLT Header)			
SIGNAL	PIN	PIN	SIGNAL
GND	2	1	GND
AUDIO LEFT	4	3	AUDIO RIGHT
AUDIO LEFT	6	5	AUDIO RIGHT
GND	8	7	GND

BOSS DAC BLOCK DIAGRAM



Software

snd-soc-allo-boss-dac.ko is the device driver required for Boss. On RPi, get this driver active by adding “dtoverlay=allo-boss-dac-pcm512x-audio” in the /boot/config.txt file.

For more details on the driver parameters & ALSA settings for Boss, please refer to our ***Technical Details For BOSS DAC*** *document.*